

College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Midterm Examination of Semester 2

Subject: 311131 Mechanical Drawing I

Date: 09 March 2016

Year: 2015

Section: 5-6

Time: 15.00-17.00

Name: _____ ID: _____ Field of Study: _____

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
 2. No documents are allowed to be taken out of the examination room.
 3. Write your name, student ID, section, and seat number clearly on the answer sheet.
 4. This is a close book test.
 5. A dictionary and a calculator are not allowed.
 6. The examination has 9 pages (including this page), 9 problems and a total score of 60 points.
 7. Write all your answer in this examination sheet.
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Problem 1 Answer all these questions

(5 marks)

- If the top view of an object shows a drilled through hole, how many hidden lines would be necessary in the front view to describe the hole?

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- List the six principal views of multiview projection.

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- In a drawing that shows the top, front, and right-side view which two views show depth? Which view shows depth vertically on the sheet? Which view shows depth horizontally on the drawing sheet?

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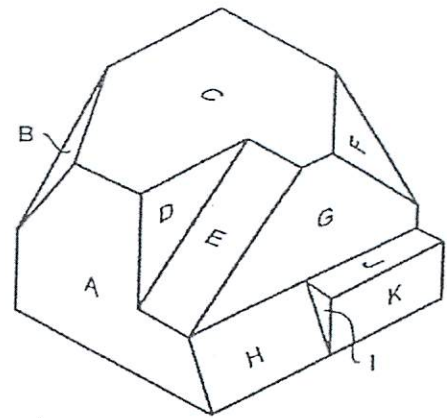
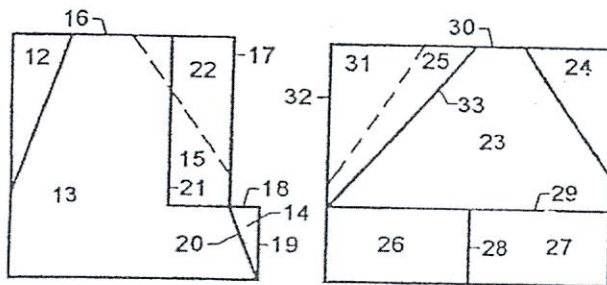
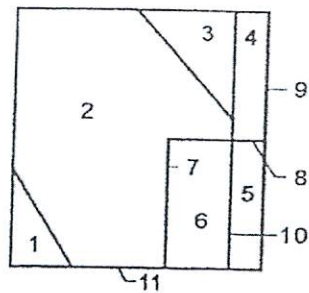
- What can the draft-person do to prevent confusion between first and third angle projection?

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- A straight line always projects as a or as a

Problem 2 From given 3rd angle orthogonal projection views and pictorial (isometric) drawing, match the surface numbers and surface A-K, as an example of surface A. (10 marks)



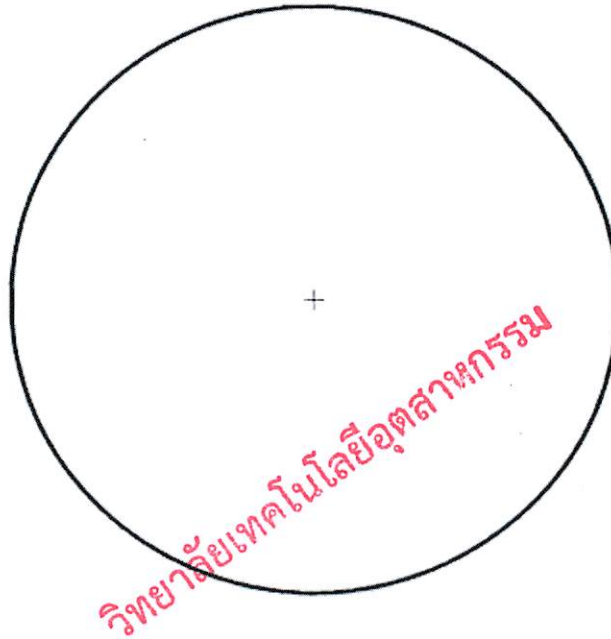
Surface	Front view	Side view	Top view
A	13	32	11
B			
C			
D			
E			
F			
G			
H			
I			
J			
K			

Instructions: (for Problem 3-5)

- The following exams must be constructed with the aid of compasses and set squares.
- Ensure to use correct and accurate methods.
- All construction lines and projection lines must be shown.
- A uniform thickness and darkness of visible line is required throughout.
- Emphasis is placed on creating drawings that are neat, correctly shape and scale.

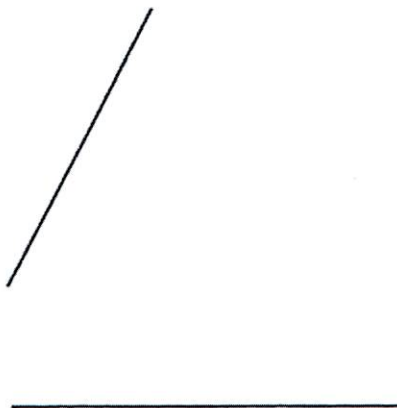
Problem 3 Construct a hexagon inside the circle.

(5 marks)

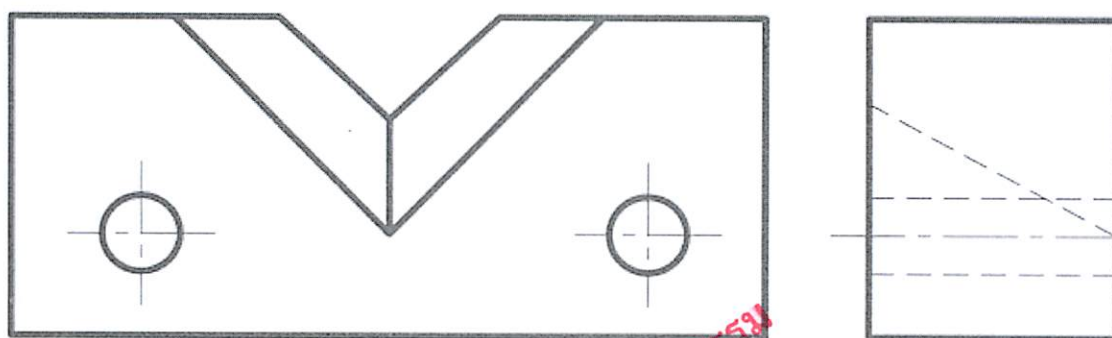


Problem 4 Construct an arc with a radius of 30 mm tangential to given lines.

(5 marks)



Problem 5 From given 1st angle projection of front and side view drawing of V-Block, complete the missing top view. (5 marks)

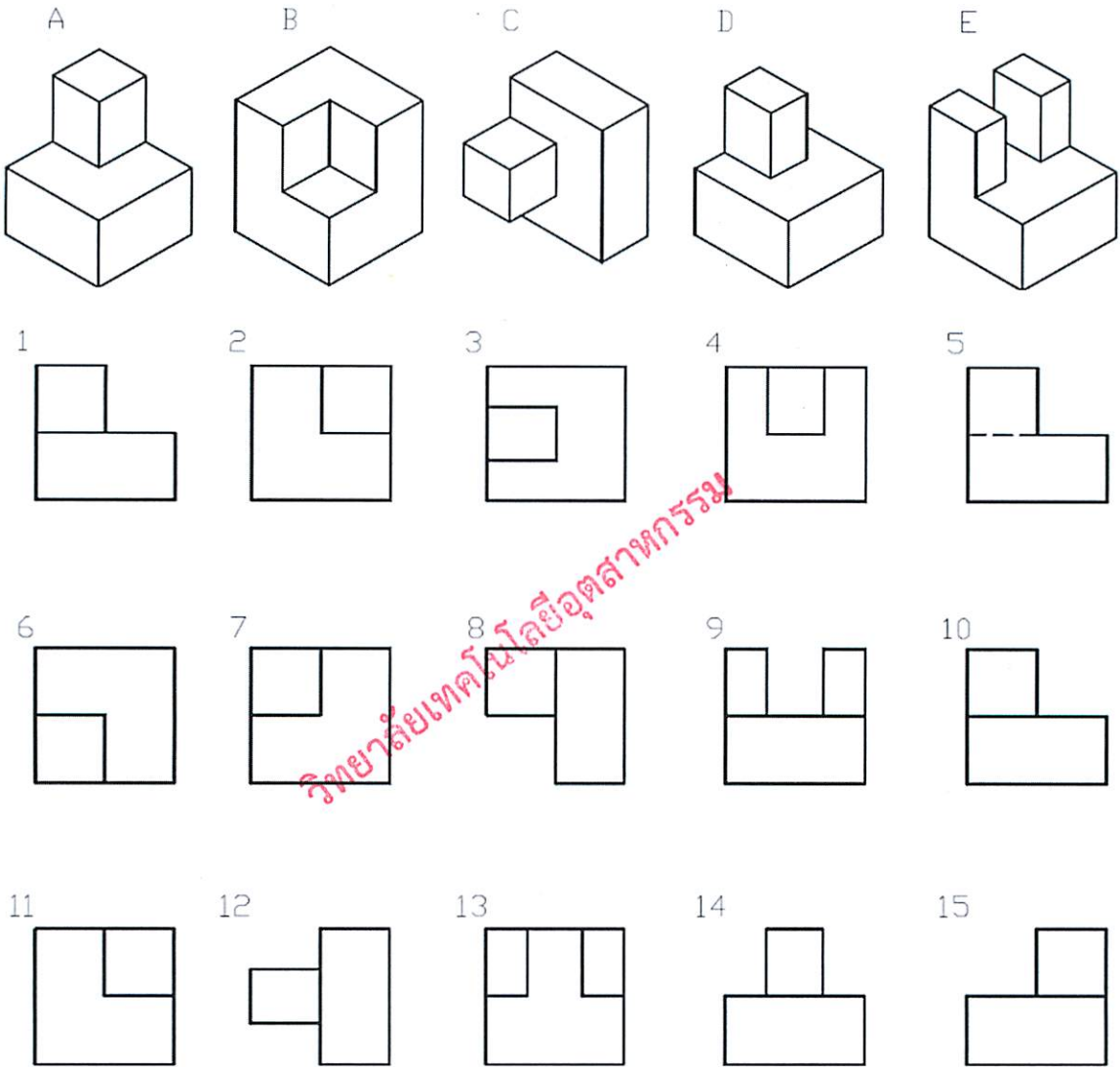


V-BLOCK

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Problem 7 From given isometric drawing, select front, top and side views.

(5 marks)



ISOMETRIC DRAWING	A	B	C	D	E
FRONT VIEW					
TOP VIEW					
SIDE VIEW					



Problem 8

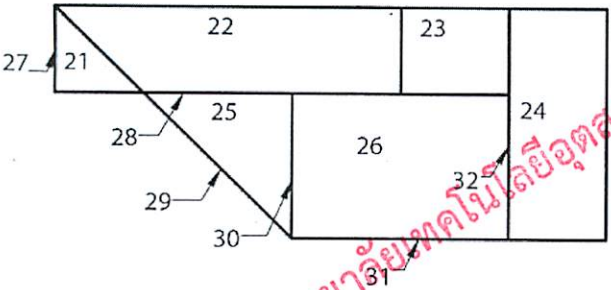
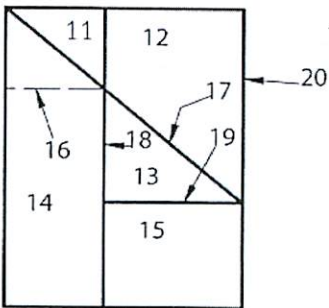
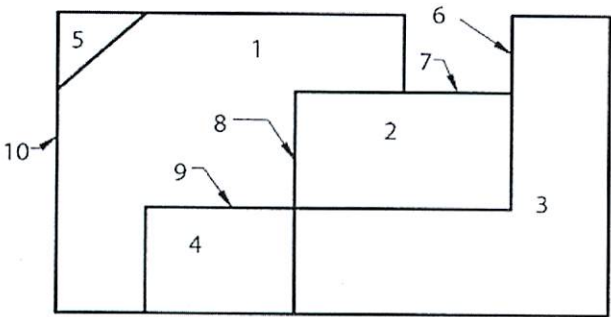
Select the multiview drawing from given isometric drawing.

(5 marks)

1		 A	 B	 C
2		 A	 B	 C
3		 A	 B	 C
4		 A	 B	 C
5		 A	 B	 C



Problem 9 From given 1st angle multiview drawing, complete the table below by matching the surface numbers from each view (front, top and side views). (10 marks)



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TOP	FRONT	SIDE